Space and Planets: A Short Reference Guide

This document is designed for testing Retrieval-Augmented Generation (RAG) systems. It contains information across multiple topics related to astronomy and planetary science.

1. The Solar System

The Solar System consists of the Sun, eight planets, dwarf planets, moons, asteroids, and comets. The Sun contains over 99 percent of the system's mass and provides the energy that drives planetary climates.

2. Inner Planets

Mercury is the closest planet to the Sun and has almost no atmosphere. Venus has a dense carbon dioxide atmosphere that creates an extreme greenhouse effect, making it the hottest planet. Earth is the only known planet that supports life. Mars is called the Red Planet because of iron oxide on its surface and contains Olympus Mons, the largest volcano in the Solar System.

3. Outer Planets

Jupiter is the largest planet and is famous for the Great Red Spot, a giant storm that has existed for centuries. Saturn is recognized for its spectacular ring system made mostly of ice particles. Uranus rotates on its side with an axial tilt of about 98 degrees. Neptune is the windiest planet, with wind speeds exceeding 2,000 kilometers per hour.

4. Dwarf Planets

Pluto was reclassified as a dwarf planet in 2006. Other well-known dwarf planets include Eris, Haumea, Makemake, and Ceres.

5. Moons

Earth has one natural satellite, the Moon. Jupiter's moon Europa is believed to contain a liquid ocean beneath its icy crust. Saturn's moon Titan has lakes and rivers of liquid methane and ethane.

6. Space Exploration

The Voyager 1 spacecraft is the most distant human-made object and has entered interstellar space. The James Webb Space Telescope studies distant galaxies and the atmospheres of exoplanets using infrared observations.

7. Exoplanets

An exoplanet is a planet that orbits a star outside our Solar System. Thousands have been discovered using methods such as the transit method and radial velocity method. Some exoplanets lie within a star's habitable zone where liquid water could potentially exist.

8. Black Holes

A black hole is a region of spacetime where gravity is so strong that even light cannot escape. The Event Horizon Telescope produced the first image of a black hole in 2019.

9. Interesting Facts

A day on Venus is longer than a Venusian year. Mars has the largest dust storms in the Solar System. Jupiter has more than 90 confirmed moons. Saturn is less dense than water, meaning it

would theoretically float in a sufficiently large ocean.